MIL-S-19500/183A
12 February 1965
SUPERSEDING
MIL-S-19500/183 (EL)
26 June 1963

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, RF MIXER

TYPE JAN-1N25WA

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a double-ended, silicon semiconductor diode for use as a mixer (first detector) in the L-band.

1.2 Ratings.

	Type	Z(IF)	L_c	NF_o	VSWR
		ohms	db	db	ratio
Minimum	1N25WA	100	---	---	---
Maximum	1N25WA	300	6.0	8.7	2.0

OPERATING AMBIENT TEMPERATURE: -65° to + 70° C
STORAGE TEMPERATURE: -65° to + 70° C

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

Military

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Armed Services Electro-Standards Agency

114-JAN Resistor Cartridge, Test; For Crystal Diode Types 1N21, 1N23, 1N25, 1N26 and 1N78
231-JAN Equipment, Burnout Test; For Crystal Diode Type 1N25
232-JAN Equipment, Test Noise Measurement; For Crystal Diode Type 1N25
233-JAN Equipment, Test, Loss Measurement; For Crystal diode Type 1N25 (Schematic)

FSC 5960

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234-JAN Equipment, Test, Loss Measurement; For Crystal Diode Type 1N25 (Assembly)
235-JAN Equipment, Test, Noise Measurement; For Crystal Diode Type 1N25
236-JAN Equipment, Test, Burnout Test; For Crystal Diode Type 1N25
238-JAN Resistor Cartridge, Test; For Crystal Diode Type 1N25
239-JAN Diagram, Schematic, Test, Scope Interconnection; For Crystal Diode Type 1N25

Electronics Command

SC-DL-208494 Test Holder For Semiconductor Diode 1N25A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500, and as follows:

e_o ---------- Open-circuit voltage of the pulse generator
R_c ---------- Resistor cartridge
Z (IF) ---------- Intermediate frequency (IF) impedance

3.3 Design and construction. The semiconductor diode and adapter shall be of the design, construction, and physical dimensions as specified in figures 1 and 2. The removable adapter as shown in figure 2 shall be attached to one end of the diode body.

3.3.1 Plating. The diode shall be plated as specified in figures 1 and 2.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The marking shall be placed on each device in accordance with MIL-S-19500.

3.5.1 The manufacturer, at his option, may omit the following markings:

(a) Country of origin.
(b) Manufacturer's identification.

3.6 Burnout by repetitive pulsing. At the end of all manufacturing processes and prior to selecting samples for testing, all diodes shall be subjected to 100 percent burnout by single pulse test, only once, and shall be performed in accordance with Method 4146 of MIL-STD-750, Dwg 231-JAN, Dwg 236-JAN, Dwg 238-JAN, Dwg 239-JAN, e_o = 12 ±0.25 v, t_p = 1.0 μsec, R_{c1} = 3 ohms and R_{c2} = 12 ohms.

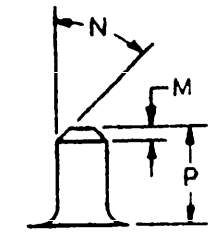
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

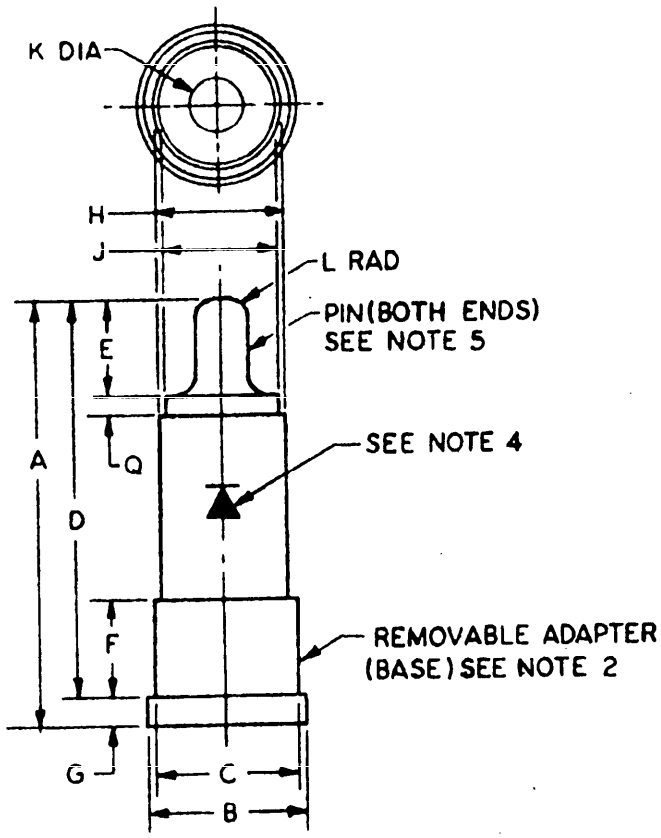
4.2 Test conditions. Unless otherwise specified herein, the test conditions, when applicable, shall be as follows:

P = 1.0 mW ± 5%
f = 1,000 ± 10 mc
Z_m = 200 ± 10 ohms; ± jo ohms
R_L = 100 ± 1 ohms

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ALTERNATE CONSTRUCTION OF PIN

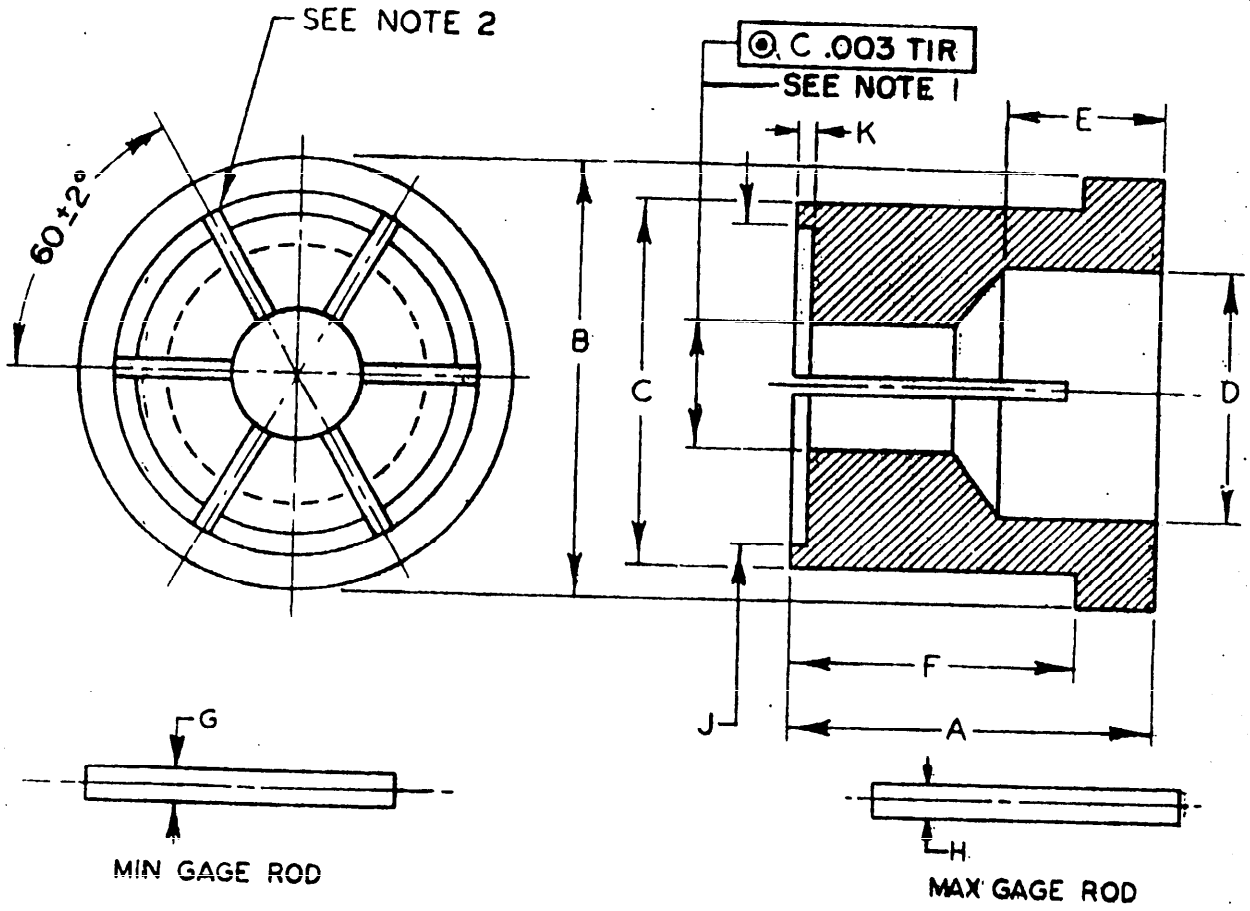


DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.800	.840	20.32	21.33
B	.292	.296	7.42	7.51
C	.246	.250	6.25	6.35
D	.753	.783	19.13	19.88
E	.180	.190	4.58	4.82
F	.193	.199	4.91	5.05
G	.047	.057	1.20	1.44
H		.240		6.09
J	.195	.215	4.96	5.46
K	.092	.094	2.34	2.38
L	.030	.046	.77	1.16
M	.010	.030	.26	.76
N	10°	45°		
P	.180	.190	4.58	4.82
Q		.030		.76

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Eccentricity between pin and base shall not exceed .0075 (.19 mm).
3. Metal parts shall be silver plated 20 milligrams per square inch (msi) minimum or gold plated 10 msi minimum.
4. The polarity shall be indicated by an arrow, with the arrow pointing in the direction of easier current flow.
5. Both pins must have convex ends.

Figure 1. Semiconductor device, diode, type 1N25WA.

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NOTES:

1. This diameter should be .095 (2.41 mm) min, .097 (2.46 mm) max diameter before closing jaws. This diameter shall then be sufficiently closed, and the adapter so tempered that it will fit on the minimum gage rod and also on the maximum gage rod, in each case with a snug fit (for hand assembly).
2. Each slot shall be .008 (.20 mm) min, .012 (.30 mm) max wide by .185 (4.70 mm) deep before closing. The 6 slots are equally spaced.
3. The material for the adapter shall be beryllium copper and the finish is gold plated 10 msi minimum.
4. All burrs and sharp edges shall be removed.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN.	MAX.	MIN	MAX
A	.246	.254	6.25	6.35
B	.292	.296	7.42	7.51
C	.246	.250	6.25	6.35
D		.187		4.75
E		.100		2.54
F	.195	.199	4.96	5.06
G	.0915	.0920	2.324	2.336
H	.0940	.0945	2.387	2.400
J	.216	.221	5.48	5.61
K	.012	.018	.30	.49

Figure 2. Removable adapter included as a part of type 1N25WA.

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4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A and B.

4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II, and as follows:

4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overall noise figure, shall be raised to $70\ ^{+5}_{-0}$° C min and maintained at this temperature until equilibrium is reached. the NF_o shall then be determined and shall be less than 11.7 db. The temperature shall then be returned to 25° C ±3° C at which time NF_o shall be less than 9.7 db.

4.5.2 Microwave parameters. The L_c, NR_o, and NF_o parameters may be determined by any suitable combination of measured parameters selected from among the following: L_c, NR_o, NF_{if} (actual), and NF_o (actual). A measurement of either L_c or NR_o, but not both, is required. NF_o shall be determined for an assumed or actual intermediate-frequency noise figure (NF_{if}) of 1.5 ± 0.25 db.

4.5.3 Group-burnout by repetitive pulsing. The diode shall be subjected to pulses as specified in subgroup 3 of group B inspection, and as follows:

(a) The patterns obtained on the oscilloscope, when R_{c1} and R_{c2} are inserted in the diode test position, shall lie within the envelope formed by these patterns and a line drawn parallel to the time axis at 20 percent of the maximum amplitude of the pattern of R_{c2}.

(b) At the completion of each test, the diode sample shall be measured for conversion loss and output noise ratio.

(c) The diode samples shall again be subjected to the test procedure specified in subgroup 3 of group B inspection except that the condition for pulse voltage shall be 25 v.

(d) The diode samples shall be re-measured for conversion loss and output noise ratio. The diode shall be considered as having failed the test for group-burnout by repetitive pulsing if the conversion loss or noise ratio is degraded by more than 2.0 db or 2.0 times, respectively.

Table I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			10				
Voltage standing wave ratio	4136	Dwg SC-DL-208494		VSWR	---	2.0	---
Overall noise figure (see 4.5.2)	4126	Test cond. A		NF_o	---	8.7	db

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Table I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 - Continued							
Conversion loss (see 4.5.2)	4101	Dwgs 233-JAN and 234-JAN		L_c	---	6.0	db
IF impedance	4116	Dwgs 233-JAN and 234-JAN		Z(IF)	100	300	ohms
Output noise ratio (see 4.5.2)	4121	Dwgs 114-JAN, 232-JAN, and 235-JAN; R_c = 200 ohm		NR_o	---	1.5	---

Table II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1							
Physical dimensions	2066	Dim. A, C, D and F of figure 1	10	---	---	---	---
		Dim. B and J of figure 1 (Dim E, G, K, and note 3 of figure 1 and note 3 of figure 2 are for qualification only.)	20	---	---	---	---
Subgroup 2			20				
Thermal shock (temperature cycling)	1051	Test cond. A $T(high) = +70\ ^{+5}_{-0}\ ^{\circ}C$		---	---	---	---
Moisture resistance	1021	No initial conditioning		---	---	---	---
End points: Overall noise figure (see 4.5.2)	4126	Test cond. A		NF_o	---	9.7	db
Subgroup 3			20				
Group-burnout by repetitive pulsing (see 4.5.3)	4141	Dwgs 231-JAN, 236-JAN, 238-JAN and 239-JAN; e_o = 12.5 ± 0.25 v; tp = 1.0 µsec; R_{c1} = 3 ohms; R_{c2} = 12 ohms		---	---	---	---
Subgroup 4			20				
Shock	2016	500 G; t ≅ 1 msec 5 blows each orientation: X_1, Y_1, and Y_2		---	---	---	---

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Table II. Group B inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 4 - Continued							
Vibration, variable frequency	2056	15 G; X_1, Y_1, and Y_2 orientations		---	---	---	---
Constant acceleration	2006	10,000 G; X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as for subgroup 2)							
Subgroup 5			20				
High temperature operation (see 4.5.1)		$T_A = 70 \, ^{+5}_{-0}$°C		---	---	---	---
Subgroup 6			$\lambda = 15$				
High temperature life (nonoperating)	1031	$T_A = 70 \, ^{+5}_{-0}$°C		---	---	---	---
End points: (Same as for subgroup 2)							

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall conform to MIL-S-19500.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Preparing activity:
Army - EL

(Project 5960-1999)

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - EL, MU, MI
Navy - WP, MC, CG
Air Force - None

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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